ABSTRACT OF THE DISCLOSURE

A communication protocol that provides forward secrecy at the physical encoding layer by periodically rotating the data encoding map used during a communication session. The method comprises: initiating a session with a first disposable encoding map; monitoring a trigger condition, such as the number of transmitted data frames or elapsed time; upon the trigger condition being met, autonomously generating and switching to a new, second disposable encoding map mid-session; and continuing the communication. Each map is derived from a shared secret, but a compromise of the secret at one point in time does not allow for the decryption of traffic from an earlier epoch that was encoded with a since-discarded map. This prevents retrospective cryptanalysis of captured data streams.